BREEZE: Training-Free and Auditable LLM Recipe Augmentation with Physics-Guided Admission

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ABSTRACT

Training a target-specific waveform generator is often impractical when only a few labelled faults are available. We present BREEZE, a training-free augmentation framework in which an LLM selects structured recipes from supplied kinematics, training-fold statistics, and exemplar descriptions, and a fixed seeded renderer constructs the waveforms. A physical verifier calibrated only on the outer-training split admits candidates, returns bounded feedback, and archives recipes, seeds, hashes, and gate evidence without generator optimization or LLM fine-tuning. Across frozen, leakage-resistant few-shot protocols on three public datasets, the resulting fixed pools yield stable paired improvements over repeated subset draws and CNN initializations: PU LLM recipes exceed rule and random open-loop sources at 5, 10, and 25 shots, while CWRU improvements over rule, noise augmentation, and real-only hold within load0 and from source load0 to held-out loads 1–3 under family-wise Holm correction. On case/run-disjoint Berkeley milling, LLM recipes exceed both non-structured baselines at every shot; their significant advantage over the structured rule is confined to lower-shot cells, is only 0.2–0.5 percentage points, and converges at ten shots. A global Benjamini–Hochberg sensitivity analysis preserves every registered decision, whereas six predeclared PU cross-condition attempts delimit the current morphology-transfer boundary. These results identify auditable, training-free admission as a practical augmentation operating point for the stated few-shot protocols.

1. Introduction

Industrial fault diagnosis faces a structural data imbalance. Healthy operation is abundant, whereas labelled faults are scarce, expensive to seed, and rarely observed under every load and speed. New machines, sensor layouts, and operating regimes therefore offer only a few examples of the events that a diagnostic model must recognize.

Target-trained augmentation exposes concrete failure mechanisms under scarce or imbalanced faults. DFE-GAN addresses adversarial instability, mode collapse, and weak sample filtering through a modified objective, attention, and an automatic filter (Liu, Jiang, Wu and Li, 2022). Diffusion removes the adversarial game but retains target distribution fitting: TSDM redesigns the original DDPM denoiser for one-dimensional vibration structure, ReF-DDPM strengthens intra- and inter-level features, a TII study adds a comprehensive evaluation framework, and FaultDiffusion attributes few-shot failure to the normal–fault domain gap and high intra-class variation (Yi, Hou, Jin, Saeed, Kandil and Duan, 2024; Yu, Li, Huang, Xiao, Zhang, Li and Fu, 2024; Yang, Ye, Yuan, Zhu, Mei and Zhou, 2024; Xu, Chen, Wang, Li, Tang, Zhao and Liu, 2025). ACS-DM and MR-DDIM further modify trained denoisers and objectives to balance temporal diversity with spectral fidelity (Tong, Zhu, Lu, Wang, Du, Jiang, Xu, Huo and Du, 2026; Xu, Tong, Zhu, Du, Zhao, Jiang and Wang, 2026).

These methods share an operational requirement: the same limited target corpus must support distribution fitting, design choices, and checkpoint selection. Even when impact priors or spectral constraints enter the objective, they influence generated signals through target-data optimization and model selection (Gao, Xu, Zhang, Fang and Tan, 2025; Liu, Han, Chen, Ding, Zhang, Shen, Wu, Zhang and Li, 2026). We study a different operating point in which target-specific generator fitting is removed and the controllable signal description remains explicit.

BREEZE changes where controllable generation choices reside. First, the LLM selects kinematics, impacts, modulation, and spectral allocation in a structured JSON recipe using prompt-supplied machine information, training-fold statistics, and exemplar descriptions; a fixed seeded renderer executes those fields as explicit signal equations. The evidence therefore tests LLM-mediated recipe selection, rather than whether pretraining alone contains fault physics. Second, a verifier moves physical control from a learned objective to gates calibrated on the corresponding outer-training split, and measured failures condition bounded resampling. Third, recipe, seed, waveform hash, and gate report records make every admitted sample traceable. Figure 1 summarizes these components.

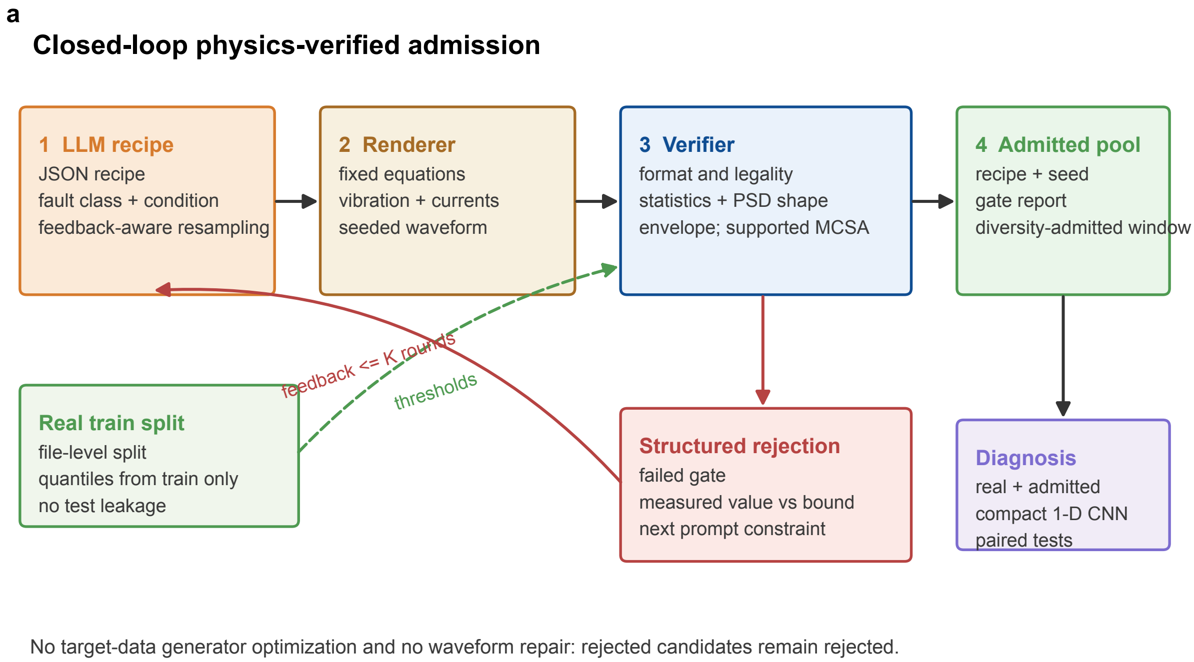

Figure 1: The BREEZE workflow. The LLM proposes a structured recipe, a fixed seeded renderer constructs vibration and, when available, current channels, and a verifier calibrated exclusively on the real training split admits or rejects the candidate. A rejection becomes structured feedback for at most K further rounds. Rejected signals are not repaired.

Three evaluation questions test this causal chain. First, can random legal values or a fixed engineer-written rule replace the LLM's recipe knowledge when renderer and verifier are held constant? Second, how does train-calibrated closed-loop admission shape feasible pools and their physical evidence? Third, do the resulting pools improve file-, load-, or experiment-disjoint few-shot diagnosis, and where does LLM-mediated recipe selection stop transferring? Paired source ablations, admission records, class-conditional diagnostics, leakage-resistant splits, and corrected tests answer the questions together.

The paper makes four contributions:

1. **Externalize** prompt-supported generation choices through LLM-mediated recipe selection and a fixed seeded renderer, separating semantic proposal from waveform construction without optimizing a target-data generator.
2. **Convert** physical control from a generator-training objective to train-calibrated admission over robust statistics, Welch-PSD shape, envelope-spectrum evidence, reliability-gated current evidence, and diversity, with bounded feedback and per-sample audit records.
3. **Isolate** the LLM-mediated recipe contribution through a verifier-matched rule source and a distinct random open-loop control on three public datasets: PU, CWRU, and Berkeley milling. The registered evidence comprises 20 PU seeds and 40 seeds for CWRU and Berkeley, with family-wise Holm correction and a global BH sensitivity audit.
4. **Delineate** the operating boundary through the complete six-stage PU leave-one-condition-out (LOCO) failure chain, process-metadata confounding in UMich milling, and a stopped MU-TCM inner-validation line.

The evidence supports a deliberately bounded recipe-generation and admission claim: admitted candidates satisfy declared train-supported gates and the fixed pools improve the stated few-shot tasks under their frozen protocols. The scope is training-free augmentation, not training-free diagnosis; few-shot real calibration and a trained downstream classifier remain necessary.

2. Related work: where generative knowledge resides

2.1. Knowledge learned from target signals

General time-series generators place distributional knowledge in learned weights. Recurrent conditional GANs, TimeGAN, transformer GANs, variational models, and diffusion models all fit a target sequence corpus (Esteban, Hyland and Rätsch, 2017; Yoon, Jarrett and van der Schaar, 2019; Li, Metsis, Wang and Ngu, 2022a; Li, Ngu and Metsis, 2022b; Liao, Ni, Szpruch, Wiese, Sabate-Vidales and Xiao, 2020; Desai, Freeman, Wang and Beaver, 2021; Nichol and Dhariwal, 2021; Tashiro, Song, Song and Ermon, 2021; Alcaraz and Strodthoff, 2022; Rasul, Seward, Schuster and Vollgraf, 2021). Machinery-specific GANs improve this fitting through sparsity, feature enhancement, frequency learning, spectrum-guided sampling, and limited-data architectures (Ma, Ding, Wang, Wang, Ma and Lu, 2021; Liu et al., 2022; Ko, Lee, Kim, Kim and Youn, 2023; Kim, Ko, Lee, Kim, Jung and Youn, 2024; Lin, Kong, Zhao, Han, Dong, Liu and Chu, 2025). Recent diffusion studies replace adversarial learning with trained denoising while redesigning feature paths or sampling for vibration data (Yi et al., 2024; Yu et al., 2024; Yang et al., 2024; Xu et al., 2025; Tong et al., 2026). These methods retain knowledge in fitted generator weights; BREEZE instead evaluates an operating point without target-generator optimization.

2.2. Physical knowledge learned through objectives and architecture

Physics-informed generation adds domain knowledge to the learning process. Equations, mechanisms, or physical penalties constrain a trainable model (Raissi, Perdikaris and Karniadakis, 2019); rotating-machinery systems encode impact response, speed modulation, multimodal structure, condition embeddings, or spectral consistency (Gao et al., 2025; Liu et al., 2026; Guo, Xu, Zheng, Xie and Wang, 2025; Xu et al., 2026). The prior resides in an objective or architecture, but its effect on generated signals still passes through optimized weights. BREEZE places the signal equations in a fixed renderer and evaluates physical evidence after rendering through train-calibrated admission.

2.3. Description-conditioned learned generation

Language descriptions provide another route for conditioning a learned model. LLMs have been used as numerical time-series forecasters and supplied with machinery priors for bearing health management (Gruver, Finzi, Qiu and Wilson, 2023; Peng, Liu, Du, Gao and Wang, 2025). BearGen is the closest signal-generation study: an LLM produces a signal description that conditions a diffusion generator trained across multiple bearing datasets (Lee, Jeon, Kim and Kim, 2026). Its description enters generation through learned denoising weights; BREEZE asks for an executable numerical recipe and uses a fixed renderer whose target-data path contains no learned generator weights.

2.4. Explicit knowledge with calibrated admission

Generated-data filtering controls which fitted samples reach a classifier. DFE-GAN and SGAN-DDS show that filtering, spectrum guidance, and diversity sampling affect the retained pool (Liu et al., 2022; Kim et al., 2024). BREEZE combines an explicit recipe, fixed equations, train-only gate calibration, structured feedback, and per-sample records. Physical knowledge therefore resides in the recipe and renderer, while training data determine the supported admission region rather than generator weights.

Few-shot transfer and meta-learning address a complementary question: how a diagnostic classifier acquires or adapts representation knowledge from related tasks and source domains. Bearing and machinery studies use episodic or meta-trained adaptation under limited target labels and varying conditions (Li, Li, Zhang, He, Liao and Hu, 2021; Wu, Zhao, Sun, Yan and Chen, 2020; Su, Xiang, Hu, Xu and Yang, 2022; Feng, Chen, Xie, Zhang, Lv and Pan, 2022). BREEZE does not replace those classifier-side methods; it tests whether an auditable synthetic pool can improve the stated few-shot protocol without fitting a target-data generator.

2.5. Diagnostic observables used as gates

Established diagnostic observables expose complementary aspects of rolling-element faults. Envelope analysis, spectral kurtosis, and the fast kurtogram reveal stochastic or pseudo-cyclostationary impacts (Randall and Antoni, 2011; Antoni, 2006, 2007), while Welch PSD estimation represents spectral shape (Welch, 1967). Motor-current sidebands add bearing evidence when the real training data support class separability (Schoen, Habetler, Kamran and Bartheld, 1995). BREEZE differs by converting these observables into explicit, train-calibrated admission gates.

Table 1
Positioning by where generative and physical knowledge resides. "Target training" denotes optimization of a generator on the target signal corpus.

Method family	Generative mechanism	Target training	Where physical knowledge resides	Physical control	Primary evidence pattern
DFE-GAN / SGAN-DDS	Learned adversarial generator plus filtering or sampling	Yes	Generator and discriminator weights	Learned features, filtering, and spectrum guidance	Fidelity, diversity, and diagnosis
Physics-informed GAN / diffusion	Learned generator with physical priors or condition embedding	Yes	Physical objective, architecture, and optimized weights	Training loss or architecture	Mechanism, multidomain fidelity, diagnosis
BearGen	LLM descriptions condition a learned diffusion model	Yes	Description prompt and learned denoising weights	Description-conditioned denoising	Eight bearing datasets, quality, few-shot, cost
BREEZE	LLM JSON recipe plus fixed seeded renderer	No	Recipe, renderer equations, and calibrated gate thresholds	Train-calibrated admission and rejection	Source ablation, gate reports, physical distances, paired diagnosis

3. Problem formulation and framework

3.1. Train-calibrated admission

The second component moves physical control out of generator training and into a decision rule calibrated entirely within the outer training split. Let $\mathcal{D}_{\mathrm{tr}} = \{(x_i, y_i, m_i)\}_{i=1}^{n}$ denote its real windows, class labels, and available machine metadata. For requested class y and condition m, the LLM proposes recipe r_t at feedback round t. The following mapping makes recipe execution and local randomness explicit:

$$\tilde{x}_{t,s} = R(r_t, m; s), \tag{1}$$

where the seeded renderer R archives s with the recipe. The verifier then maps the rendered waveform to both an admission decision and its evidence report:

$$(a_{t,s}, q_{t,s}) = V(\tilde{x}_{t,s}, y, m; \mathcal{D}_{\mathrm{tr}}), \qquad a_{t,s} \in \{0, 1\}. \tag{2}$$

Equation (2) exposes the decision $a_{t,s}$ and the diagnostic report $q_{t,s}$ as separate outputs. Renderer inputs, gate calibration, feedback, and pool selection all remain confined to the outer training split.

Admission must represent interval support, direction-specific evidence, healthy suppression, and diversity in one decision. Partition the active feature gates into two-sided, upper-bound, and lower-bound index sets $\mathcal{J}_{\mathrm{int}}$, $\mathcal{J}_{\mathrm{up}}$, and $\mathcal{J}_{\mathrm{low}}$. For deterministic feature f_j, class-supported bounds $[l_{j,y,m}, u_{j,y,m}]$, healthy fault-prominence score p_f, physical embedding $e(x)$, and current admitted set $\mathcal{S}_y$, the implemented predicate types are

$$\begin{aligned}
l_{j,y,m} \le f_j(x) \le u_{j,y,m}, \quad j \in \mathcal{J}_{\mathrm{int}} \quad & \text{(interval)},\\
f_j(x) \le u_{j,y,m}, \quad j \in \mathcal{J}_{\mathrm{up}} \quad & \text{(upper-bound evidence)},\\
f_j(x) \ge l_{j,y,m}, \quad j \in \mathcal{J}_{\mathrm{low}} \quad & \text{(lower-bound evidence)},\\
p_f(x) \le u_{f,\mathrm{healthy}} \quad & \text{(healthy suppression)},\\
\mathcal{S}_y = \varnothing \text{ or } \min_{z \in \mathcal{S}_y} \|e(x) - e(z)\|_2 \ge \delta_y \quad & \text{(diversity)}.
\end{aligned} \tag{3}$$

The empty-set clause admits the first candidate with respect to diversity only; all active per-candidate gates must still pass. Subsequent candidates must meet the distance lower bound. The intersection of the active predicates defines the admissible set. Membership therefore records compliance with the configured gates under their training calibration.

3.2. Responsibility boundaries

A component claim is meaningful only when proposal knowledge, waveform construction, admission, and diagnosis have explicit responsibility boundaries. Figure 2 assigns one component to each responsibility. The LLM

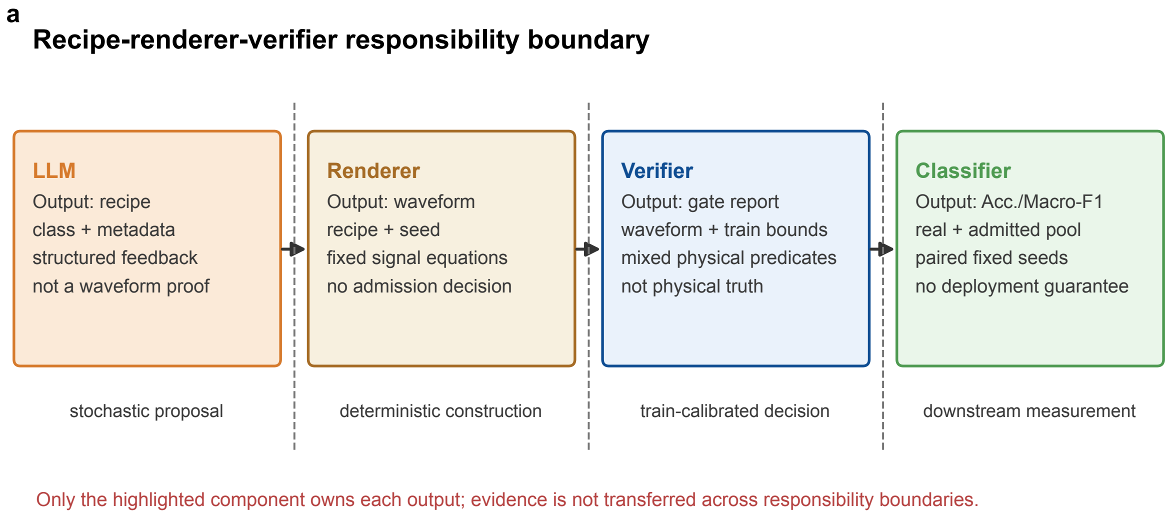

Figure 2: Component responsibility boundary. The LLM selects recipe parameters; the renderer alone constructs waveforms; the verifier admits or rejects without repair; and the classifier is used only for downstream evaluation. Every formal source comparison holds renderer and classifier fixed.

selects recipe fields, the renderer enforces the signal equations, the verifier measures compatibility with train-supported evidence, and the downstream classifier measures utility. Controlled source comparisons can then isolate the recipe source while holding the other components fixed.

The structured recipe exposes every LLM-controlled degree of freedom. It records shaft and fault frequencies, target RMS, periodic-impact rate, amplitude, decay, resonance, slip jitter, amplitude variation, load-zone modulation, background components, random machine transients, eight vibration-band weights, and optional two-phase current parameters. The initial prompt supplies class, condition metadata, bearing kinematics, a training-derived exemplar description, and the JSON schema. The feedback prompt adds the previous recipe, each measured gate violation, its supported bound, and a request for proportional parameter change. Complete prompt builders and schemas are preserved in `src/llm.py`.

3.3. Seeded renderer

The first component can bypass learned generator weights only when every recipe field enters the waveform through fixed equations. For PU vibration, the renderer combines deterministic background harmonics, Poisson machine transients, a jittered train of resonant impacts, and band-shaped noise. The equations expose the source terms and the link between impact interval and requested fault frequency:

$$x_v(t) = \sum_q A_q \sin(2\pi f_q t + \phi_q) + \sum_\ell b_\ell h(t - u_\ell) + \sum_k a_k h(t - t_k) + \epsilon_b(t), \tag{4}$$

$$h(\tau) = \mathbf{1}_{\tau \geq 0} \exp(-\tau/\tau_d) \sin(2\pi f_r \tau + \phi), \tag{5}$$

$$t_{k+1} - t_k = f_f^{-1}(1 + \xi_k). \tag{6}$$

Here u_ℓ follows a Poisson background process, f_f is the requested fault frequency, and ξ_k is bounded recipe jitter. Inner-race impact amplitudes may be modulated at shaft frequency. The renderer shapes additive noise over the fixed eight-band grid and applies the same deterministic target- RMS normalization to every recipe source before admission. Both operations belong to R and precede the gate decision.

Algorithm 1 BREEZE recipe admission for one proposal slot

Require: class y, condition m, LLM G, renderer R, verifier V, round limit K, renderer seeds $\mathcal{S}$

$q_{-1} \leftarrow \emptyset$
for $t = 0$ to K **do**
    $r_t \leftarrow G(y, m, q_{t-1})$
    for all $s \in \mathcal{S}$ **do**
        $\tilde{x}_{t,s} \leftarrow R(r_t, m; s)$
        $(a_{t,s}, q_{t,s}) \leftarrow V(\tilde{x}_{t,s}, y, m)$
        if $a_{t,s} = 1$ **then**
            archive recipe, seed, waveform hash, and full gate report
            return admitted slot and all diversity-admitted windows
        end if
    end for
    $q_t \leftarrow \text{AggregateFailures}(\{q_{t,s}\})$
end for
return rejected slot with terminal failure report

When two phase currents are available, the following equation makes carrier, fault modulation, harmonics, and channel noise explicit for channel p:

$$i_p(t) = A_p\left[1 + \beta\cos(2\pi f_f t + \psi)\right]\left[\sin(2\pi f_e t + \phi_p) + \sum_{h\in\{3,5,7\}} \rho_h \sin(2\pi h f_e t + \phi_{p,h})\right] + \eta_p(t), \tag{7}$$

where f_e is supply frequency. The code and archived seed make $R(r, m; s)$ deterministic.

Berkeley requires a dataset-specific milling renderer under the same component responsibility principle. Its multichannel renderer starts from a train-only channel exemplar, adds the recipe's transient and band-energy structure, and preserves channel-specific location and scale. This renderer is calibrated inside the outer training fold, while Eqs. (4)–(7) define the bearing-channel model.

3.4. Verifier gates

The second component moves physical constraints from a generator-training loss to calibrated gates, so the verifier evaluates six complementary evidence families:

1. **Legality:** exact shape, finite samples, non-degenerate channels, and absence of long repeated runs.
2. **Robust statistics:** class- and channel-conditional RMS, standard deviation, peak, kurtosis, skewness, and crest factor inside train-derived quantile or robust-union domains.
3. **Spectral shape:** triangular-band Welch-PSD fractions and PSD-CDF Wasserstein-1 distance to the class training reference.
4. **Envelope evidence:** resonance-band Hilbert demodulation, prominence near the requested BPFO, BPFI, BSF, or FTF frequency, harmonic support, and suppression of unsupported peaks for healthy candidates.
5. **Current evidence:** vector-current sideband scores become active when training data establish class separability; otherwise they remain audit-only measurements.
6. **Diversity:** a lower nearest-neighbour bound in the physical feature embedding relative to a train-derived real–real reference.

PU training data leave the MCSA score audit-only. Current-signature consistency is therefore reported only for pools whose training split activates that gate.

3.5. Closed-loop construction and audit records

Measured failures shape the pool only when admission closes the loop around the proposal source. This loop must bound proposal cost while preserving every decision needed for audit and replay. Algorithm 1 states the contract for one proposal slot.

Table 2
Formal datasets and frozen protocols. Counts are windows after the declared split.

Dataset	Sampling and input	Classes	Operating regime	Train/test	Split unit
PU	8 kHz, 3 channels, 2048 samples	healthy, OR, IR	N09_M07_F10: 900 rpm, 0.7 Nm, 1000 N	3846/962	chronological measurement files within bearing; zero shared file/window keys
CWRU	12 kHz drive-end vibration, 2048 samples	healthy, IR, ball, OR	motor loads 0–3; defects 0.007–0.028 in	696/306 for within-load0	chronological within-file for load0; loads 1–3 held out separately for clean transfer tests
Berkeley	250 Hz, 6 channels, 512 samples	healthy ($VB < 0.2$), degraded ($VB \geq 0.2$)	feed, depth, and material combinations	1836/646	disjoint machining case/run groups

A *slot* is one requested class-conditioned recipe unit, whereas a *window* is the $C \times N$ tensor supplied to the classifier. One slot can archive candidates from several feedback rounds, and its admitted recipe can be rendered with several deterministic seeds. The frozen pipeline renders three seeds per recipe and can retain additional diversity-admitted expansions, so one admitted slot can yield multiple windows. This distinction matters because slot counts measure LLM proposal cost, while window counts measure the classifier's augmentation budget.

Generation provenance determines which stages can be replayed exactly. The CWRU archive records the OpenAI-compatible model identifier `mimo-v2.5`, temperature 0.8, top-p 0.9, and maximum 900 output tokens, although the provider did not expose sampling seeds. The earlier PU archive preserves prompts, recipes, rendered pools, and local seeds; its original provenance record omits the exact provider version and top-p. Consequently, the frozen PU pools are reproducible from archived recipes and renderer seeds, while exact regeneration of the original language responses remains unavailable. Manuscript assembly required no new LLM call.

4. Experimental design

4.1. Public datasets and leakage-resistant splits

Leakage-resistant evaluation requires splitting acquisition units before sampling windows. Window-level random splits can mix recording-specific or overlapping signal structure across partitions and thereby answer a weaker generalization question; a recent bearing analysis documents materially split-dependent conclusions, while a broader review classifies the resulting evaluation risks (Wheat, Mohrenschildt, Habibi and Al-Ani, 2024; Kapoor and Narayanan, 2023). The formal evidence uses the established PU and CWRU bearing benchmarks (Lessmeier, Kimotho, Zimmer and Sextro, 2016; Smith and Randall, 2015) and Berkeley milling from the NASA prognostics repository (Agogino and Goebel, 2007). Their file-, load-, or case/run-level protocols appear in Table 2, and only these grouped splits support formal claims.

The PU protocol isolates measurement files within each bearing. Its healthy, OR, and IR classes contain 1200, 1202, and 1444 training windows, together with 300, 301, and 361 test windows. The input retains one vibration and two phase-current channels.

The CWRU protocol tests both within-load diagnosis and source-load transfer. It evaluates a within-load0 split and three folds that hold out load1, load2, or load3. The frozen synthetic pools derive from load0, which belongs to the training side of these three folds. Reusing these pools gives the archived held-out-load0 fold target-load provenance, so that fold remains an audit artifact.

The Berkeley protocol isolates machining case/run groups before windowing. Its six channels contain spindle AC and DC current, table and spindle vibration, and table and spindle acoustic emission. Train-only inner validation fixed the healthy/degraded threshold and split before the one-time 40-seed formal run.

4.2. Recipe-source and augmentation controls

Testing whether LLM-mediated recipe selection adds value without target-generator learning requires changing the proposal source while holding every downstream factor fixed. The central ablation therefore retains the applicable renderer, data split, synthetic budget, CNN, and paired seeds:

- **LLM + verifier:** class- and condition-aware JSON recipe, physical admission, and bounded feedback.
- **Rule + verifier:** engineer-written class template with train-supported parameter ranges, the same renderer, and the same verifier; no LLM call.
- **Random open-loop:** random legal recipe values rendered without feedback. On PU this is an open-loop non-semantic downstream control; a separate random-recipe verifier audit is also reported.
- **Noise augmentation:** real few-shot windows with channel-scaled jitter and amplitude scaling, using the same synthetic count.
- **Real only:** the paired few-shot subset without synthetic data.

The completed formal families match these controls to each dataset. CWRU compares LLM with rule, noise, and real only over within-load0 and held-out loads 1–3. Berkeley adds random open-loop, while PU Phase-A focuses on LLM versus rule and random open-loop.

4.3. Few-shot classifier and statistical protocol

Matched downstream evaluation requires a fixed classifier and paired sources of randomness. All methods within a dataset therefore share the same compact 1-D CNN. Its three convolution blocks use 16, 32, and 64 channels, kernels 15, 9, and 5, stride two, max-pooling after the first two blocks, adaptive average pooling to length eight, and a linear classifier. Adam uses learning rate 3×10^{-4}, weight decay 10^{-4}, batch size 32, and cross-entropy loss. PU uses 60 epochs; the frozen CWRU and Berkeley protocols use 20. Training-channel mean and standard deviation normalize both train and test data.

The dataset-specific budgets define the registered comparison cells. PU uses $n_{\text{real}} \in \{5, 10, 25\}$, 150 synthetic windows per class, and 20 paired seeds. CWRU uses the same shot counts, 20 synthetic windows per class, and 40 seeds. Berkeley uses $n_{\text{real}} \in \{2, 5, 10\}$, 20 synthetic windows per class, and 40 seeds. Accuracy and Macro-F1 are primary, while each paired seed selects the same real subset and classifier initialization for every method in its cell.

The statistical unit is a paired downstream seed, combining a CNN initialization with a few-shot subset draw. Within each dataset protocol, every seed reuses the same fixed synthetic pool for a method; seeds are therefore not independent LLM generations or independently generated pools. The resulting inference is precisely that the pool produced by the archived generation process is stably useful across repeated few-shot selection and classifier training. It does not estimate variation across independently generated pools.

Registered inference preserves the pairing used in training. One-sided Wilcoxon signed-rank tests compare paired seeds (Wilcoxon, 1945), and Holm correction controls family-wise error within each predeclared comparison family (Holm, 1979). The registered decision follows Holm. As a cross-dataset sensitivity analysis, we additionally apply one global Benjamini–Hochberg correction to the 102 core PU, provenance-valid CWRU, and Berkeley hypotheses (Benjamini and Hochberg, 1995). Table 3 shows that this global FDR analysis preserves all 102 registered pass/fail decisions; it does not replace the preregistered family structure. We report direction, adjusted decision, seed count, and the complete pass pattern. An unadjusted p value near 0.05 remains descriptive; significance requires the adjusted Holm decision.

4.4. Physical and distribution diagnostics

Physical and distribution diagnostics complement downstream accuracy by measuring how each class relates to its outer-training reference. They include RMS, kurtosis, skewness, crest factor, peak value, BPFO/BPFI/BSF/FTF peak alignment where applicable, envelope prominence and harmonic support, PSD-CDF Wasserstein-1 distance, relative band-energy error, and nearest-neighbour distance. Wasserstein distance follows the optimal-transport interpretation of Villani (2009). The archived feature audit retains MMD with an RBF kernel and median pairwise-distance bandwidth (Gretton, Borgwardt, Rasch, Schölkopf and Smola, 2012), while the main figure reports directly interpretable distances. The core evidence uses these direct diagnostics; t-SNE, UMAP, and Mixup remain outside the frozen formal comparison.

Table 3
Multiplicity sensitivity over the 102 core paired hypotheses. Registered within-family Holm decisions remain primary; global BH applies one FDR family across PU, provenance-valid CWRU, and Berkeley.

dataset	hypotheses	Holm pass	global BH pass	agreement
PU	12	12	12	12/12
CWRU	72	72	72	72/72
Berkeley	18	15	15	18/18
total	102	99	99	102/102

Table 4
PU Phase-A v2 file-split results (20 seeds; $n_{\mathrm{syn}} = 150$ per class). Holm q values test LLM $K = 3$ superiority within each registered family.

n_{real}	metric	LLM $K = 3$	rule	random	q_{rule}	q_{random}
5	Acc.	0.777 ± 0.027	0.696 ± 0.053	0.450 ± 0.029	<0.001	<0.001
5	Macro-F1	0.776 ± 0.026	0.691 ± 0.049	0.436 ± 0.053	<0.001	<0.001
10	Acc.	0.796 ± 0.030	0.724 ± 0.042	0.483 ± 0.013	<0.001	<0.001
10	Macro-F1	0.800 ± 0.029	0.724 ± 0.040	0.488 ± 0.015	<0.001	<0.001
25	Acc.	0.818 ± 0.021	0.769 ± 0.025	0.529 ± 0.022	<0.001	<0.001
25	Macro-F1	0.821 ± 0.022	0.770 ± 0.024	0.527 ± 0.017	<0.001	<0.001

To isolate the verifier after recipe generation, a zero-API cached-candidate ablation re-screens the archived PU $K = 3$ proposals with M2 (statistics), M3 (spectral shape and PSD-W1), M4 (envelope evidence), or M5 (pool diversity) disabled one at a time. The full replay must reproduce the frozen raw and balanced pools before any variant is evaluated. Every evaluable variant retains the same $B = 150$ synthetic windows/class, file split, downstream CNN, and 20 paired seeds; a diversity threshold that cannot fill that fixed budget is reported as a capacity stop rather than resampled at a lower budget.

5. Results

The results follow the component chain established in the Introduction. PU isolates the recipe source, CWRU and Berkeley test complete generated pools, and the physical and admission audits examine how the verifier shapes those pools.

5.1. LLM-mediated recipes outperform rule and open-loop random sources on PU

The PU component ablation shows that LLM recipes outperform rule and random open-loop recipes at every registered shot count for both diagnostic metrics. All 12 pairwise tests pass Holm correction ($q < 0.001$), and Table 4 reports the paired protocol means and standard deviations.

Seed-level deltas confirm that the PU LLM advantage is positive across every registered source comparison for Accuracy and Macro-F1. Figure 3 shows these paired deltas alongside their distributions. Because the renderer and verifier remain fixed, the comparison attributes the supported PU contribution to LLM-mediated recipe selection under the supplied prompt context within the complete BREEZE pipeline.

5.2. Generated LLM pools improve few-shot diagnosis within load0 and on held-out loads 1–3

The complete load0-derived LLM pools outperform every registered comparator within load0 and on provenance-valid held-out loads 1–3. The within-load0 means exceed rule and noise augmentation at each shown shot count (Table 5), while the combined protocol yields 72/72 positive Holm-corrected LLM–comparator decisions. Figure 9 reports the source-load0 LLM–rule transfer deltas: Accuracy improves by 1.7 to 3.5 percentage points, and Macro-F1 by 1.9 to 3.6 points across shots and held-out loads 1–3.

This result establishes source-load0 cross-load diagnostic utility for the tested held-out loads 1–3 and shows where explicit recipe knowledge remains useful beyond its source load under the supplied LLM context. A strict four-fold

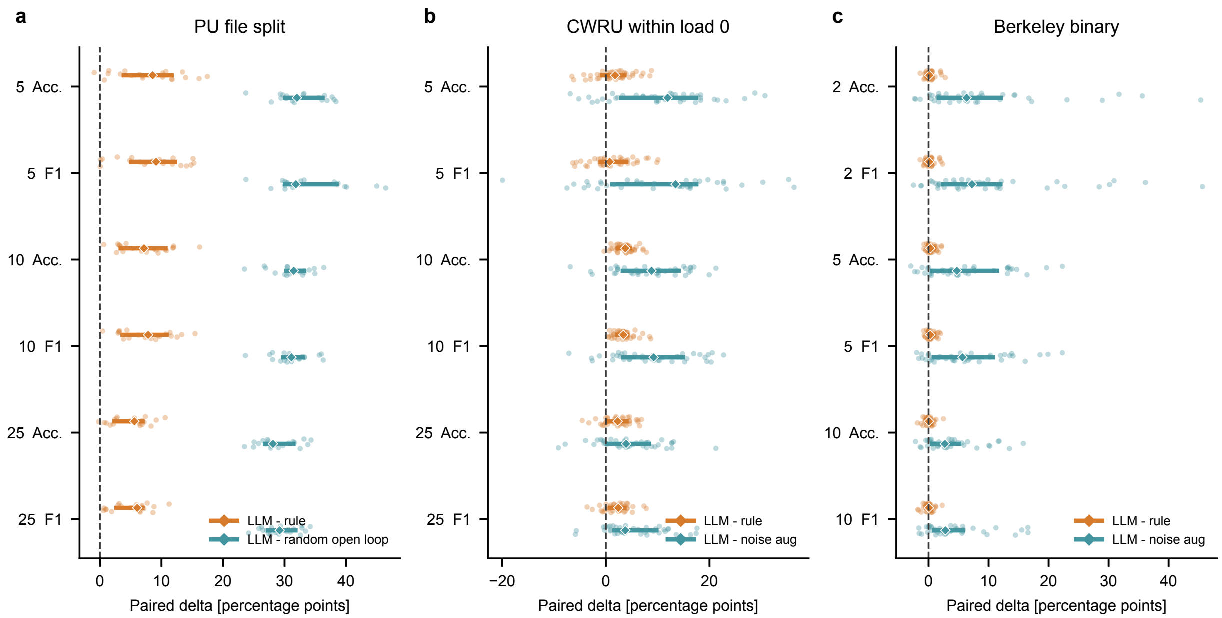

Figure 3: Paired downstream deltas. Points are seed-level differences; thick segments show interquartile ranges and diamonds show medians. (a) PU LLM minus rule or random open-loop. (b) CWRU within-load0 LLM minus rule or noise augmentation. (c) Berkeley LLM minus rule or noise augmentation. The dashed line is zero.

Table 5
CWRU within-load0 results (40 seeds). All 72/72 provenance-valid LLM-comparator tests across within-load0 and held-out loads 1–3 pass Holm correction. The archived held-out-load0 fold is excluded because it reuses a load0-derived synthetic pool.

n_{real}	metric	LLM	rule	noise augmentation
5	Acc.	0.564 ± 0.068	0.549 ± 0.053	0.454 ± 0.066
5	Macro-F1	0.567 ± 0.075	0.554 ± 0.062	0.460 ± 0.084
10	Acc.	0.621 ± 0.037	0.584 ± 0.033	0.534 ± 0.065
10	Macro-F1	0.646 ± 0.050	0.611 ± 0.046	0.554 ± 0.082
25	Acc.	0.674 ± 0.037	0.652 ± 0.037	0.628 ± 0.055
25	Macro-F1	0.711 ± 0.030	0.689 ± 0.035	0.655 ± 0.053

LOLO and cross-bearing scope remain outside this result because the reused synthetic pool gives the archived held-out-load0 fold target-load provenance.

5.3. Generated LLM pools beat non-structured Berkeley controls and converge with rules at ten shots

The complete LLM pools beat the non-structured Berkeley controls in all 12 registered comparisons. Against the strong structured rule reference, Accuracy passes at $n_{\mathrm{real}} = 2$, while the two-shot Macro-F1 comparison remains outside the Holm decision; both metrics pass at $n_{\mathrm{real}} = 5$. At $n_{\mathrm{real}} = 10$, the LLM and rule results converge and both registered comparisons fall outside the Holm decision. This complete pattern yields 15 of 18 passes, with means reported in Table 6.

The Berkeley pattern localizes the additional LLM recipe contribution to lower-shot settings, but the effect over rule is small: the passing mean deltas are 0.267 percentage points for two-shot Accuracy and 0.468/0.407 points for five-shot Accuracy/Macro-F1. At ten shots, the deltas contract to 0.055/0.020 points and do not pass Holm correction.

Table 6
Berkeley binary milling formal test (40 seeds; $n_{\mathrm{syn}} = 20$ per class). This is a partial result: all comparisons against noise augmentation and random open-loop pass, whereas LLM versus rule passes only Acc. at $n_{\mathrm{real}} = 2$ and both metrics at $n_{\mathrm{real}} = 5$.

n_{real}	metric	LLM	rule	noise aug.	random open-loop
2	Acc.	0.786 ± 0.021	0.783 ± 0.022	0.695 ± 0.110	0.524 ± 0.147
2	Macro-F1	0.778 ± 0.018	0.776 ± 0.019	0.675 ± 0.112	0.450 ± 0.127
5	Acc.	0.784 ± 0.030	0.779 ± 0.031	0.724 ± 0.068	0.614 ± 0.097
5	Macro-F1	0.773 ± 0.031	0.769 ± 0.032	0.709 ± 0.071	0.514 ± 0.106
10	Acc.	0.787 ± 0.024	0.786 ± 0.024	0.750 ± 0.049	0.654 ± 0.062
10	Macro-F1	0.777 ± 0.024	0.776 ± 0.023	0.734 ± 0.056	0.572 ± 0.081

Table 7
PU physical diagnostics for fixed 150-window/class pools. RMS, kurtosis, PSD, and band-energy entries average all three classes; alignment averages the OR and IR fault classes because it is undefined for healthy windows. Lower is better for the five reference-relative diagnostics; diversity has no preferred direction.

pool	RMS W1	kurtosis W1	PSD W1	band-energy error	alignment [Hz]	NN diversity
llm	0.008	2.658	236.9	0.044	0.929	17.10
rule	0.031	3.425	271.6	0.235	0.547	14.20
random_open_loop	0.024	2.342	516.1	0.749	0.974	52.50
noise_aug	0.004	0.274	309.1	0.014	2.544	46.09

The structured rule is therefore a strong reference, and convergence shows that structured recipe design explains most of the gain as real samples increase. This result supports only the qualified binary protocol.

5.4. Admitted PU pools preserve complementary evidence under pool-level diversity control

The verifier admits PU LLM pools that combine class-relevant waveform evidence with within-pool nearest-neighbour selection. Under the same fixed demodulation procedure, Fig. 4 compares real, LLM, rule, and random windows. Admitted LLM and rule recipes preserve impulsive and envelope structure more consistently than the random control while retaining visible stochastic variation. These traces illustrate the population-level distributions in Figs. 5–6.

The physical diagnostics do not rank LLM best on every metric. Its PU advantage relative to the two recipe-source controls is concentrated in RMS, PSD, and band-energy fidelity: Table 7 reports lower LLM band-energy error than rule and random open-loop, lower RMS distance than both, and lower PSD distance than random open-loop. Rule is closer for some PSD and fault-frequency-alignment cells, while noise augmentation has the lowest kurtosis-W1 and often the lowest RMS distance because it directly perturbs real windows. Table 8 retains every bearing kurtosis/alignment cell, including those in which rule or noise is better. The LLM nearest-neighbour value measures spacing among synthetic samples after scaling by real-reference feature variability. It is not a synthetic-to-real distance and therefore does not establish absence of training-window copying. This diversity value requires joint interpretation with fidelity because arbitrarily large distances can indicate unrealistic samples.

The separate synthetic-to-real audit in Table 9 uses the same physics-v3 pool selections and class-matched outer-training references. No available audited pool contains a byte-identical real training window, but several pools retain high maximum cross-correlation. Because no copying threshold was declared, the distances and correlations are reported as descriptive similarity diagnostics rather than proof of independence.

Berkeley uses a real train-only channel exemplar as the renderer background. Its rows therefore characterize how much carrier structure remains after the recipe-specific transient and band-energy additions; they do not support an independent-signal-generation claim.

These diagnostics test the admission component rather than rank generators in isolation. External downstream tests, controlled source comparisons, train/test separation, and failed cross-condition experiments add evidence beyond shared physical features, while gate compliance remains insufficient for physical truth.

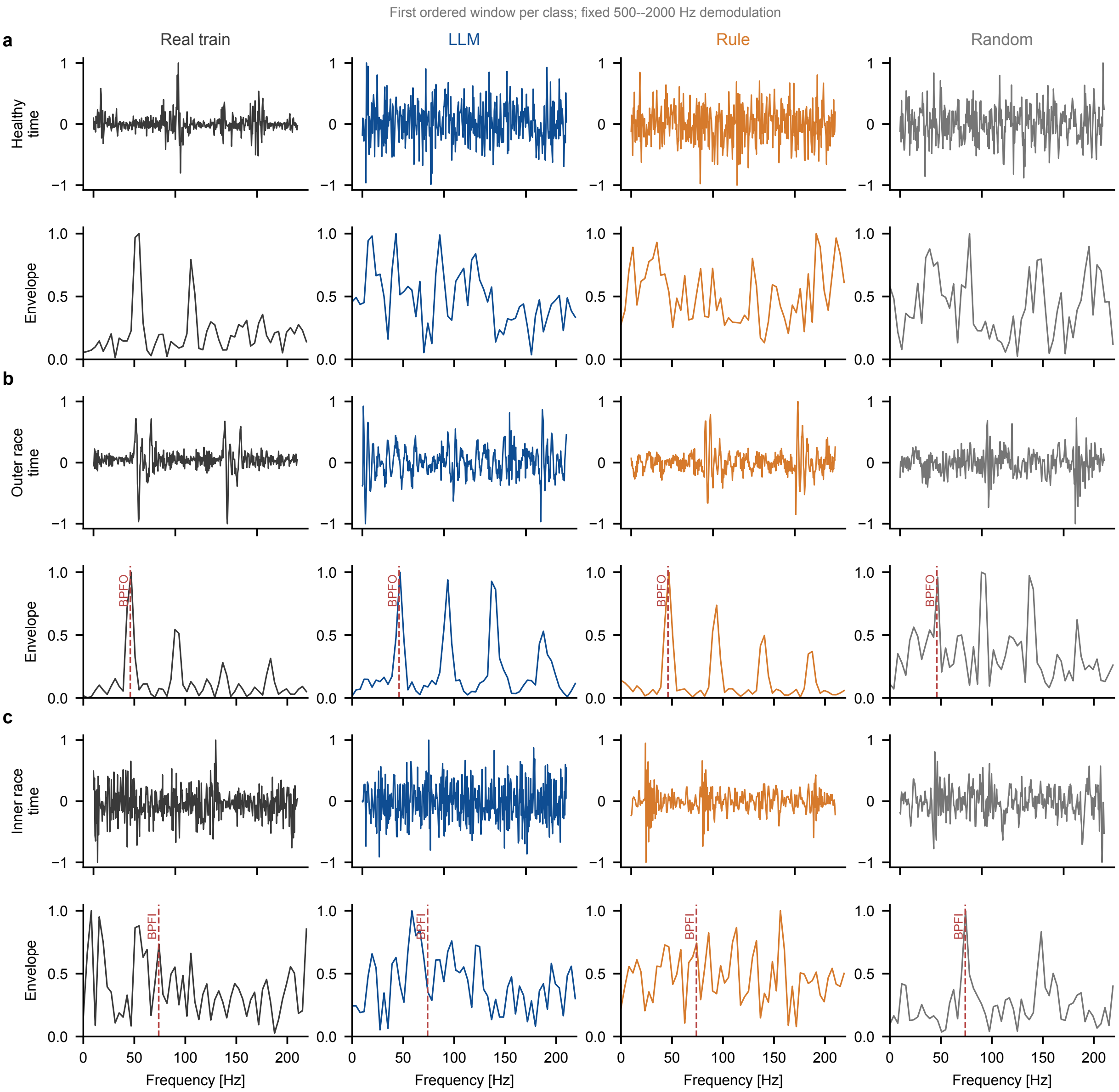

Figure 4: PU waveform and envelope-spectrum comparison. Rows are healthy, outer-race, and inner-race classes; columns compare real training, LLM, rule, and random open-loop windows. All envelope spectra use the same fixed 500–2000 Hz demodulation band. Vertical markers denote the class-relevant kinematic frequency.

5.5. Closed-loop admission shapes feasible pools and exposes failure mechanisms

The closed-loop component separates proposal cost from the downstream augmentation budget while determining which recipes can form a feasible pool. PU requests 150 proposal slots per class; Fig. 7(a) reports archived round candidates, panel (b) separates proposal slots, admitted slots, and kept rendered windows, and panel (c) compares recipe sources. At $K = 3$, the LLM process admits 60%, 60%, and 71% of healthy/OR/IR slots.

Recipe source determines whether the verifier can construct a balanced pool. The random-recipe audit admits no slot, whereas rule admission varies by class. This contrast makes the fixed rule a meaningful but incomplete substitute for LLM proposal under the PU verifier.

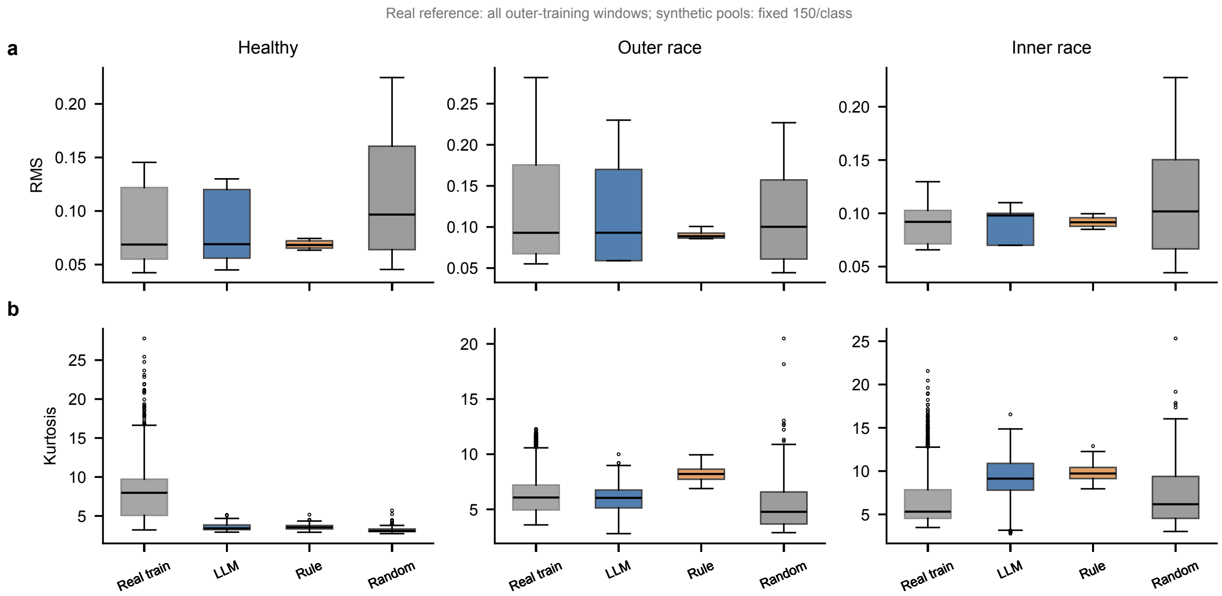

Figure 5: Class-conditional RMS and kurtosis distributions for the real PU outer-training split and frozen synthetic pools. Real boxes use every class-matched outer-training window (healthy/OR/IR: 1200/1202/1444), exactly the reference population used for the W1 diagnostics; synthetic boxes use the fixed 150-window/class pools. Admission narrows unsupported statistical excursions without forcing equality to a single reference value.

Table 8
Complete per-class bearing audit for kurtosis-W1 and fault-frequency alignment. Lower is better. Alignment is undefined for healthy windows (—). CWRU random open-loop is NA because no matching frozen physics-v3 pool exists; no value is imputed. Equal displayed values are retained rather than selectively suppressed.

dataset	pool	class	kurtosis W1	alignment [Hz]	dataset	pool	class	kurtosis W1	alignment [Hz]
PU	llm	healthy	4.495	—	CWRU	llm	B	3.881	0.466
PU	llm	OR	0.857	1.090	CWRU	llm	OR	7.865	2.055
PU	llm	IR	2.622	0.769	CWRU	rule	healthy	0.036	—
PU	rule	healthy	4.448	—	CWRU	rule	IR	5.929	2.064
PU	rule	OR	2.190	0.816	CWRU	rule	B	4.208	0.466
PU	rule	IR	3.637	0.277	CWRU	rule	OR	6.694	2.055
PU	random_open_loop	healthy	4.816	—	CWRU	noise_aug	healthy	0.043	—
PU	random_open_loop	OR	1.414	1.324	CWRU	noise_aug	IR	0.836	2.846
PU	random_open_loop	IR	0.796	0.624	CWRU	noise_aug	B	0.877	4.967
PU	noise_aug	healthy	0.371	—	CWRU	noise_aug	OR	1.516	3.555
PU	noise_aug	OR	0.229	3.043	CWRU	random_open_loop	healthy	NA	—
PU	noise_aug	IR	0.221	2.045	CWRU	random_open_loop	IR	NA	NA
CWRU	llm	healthy	0.150	—	CWRU	random_open_loop	B	NA	NA
CWRU	llm	IR	5.354	2.256	CWRU	random_open_loop	OR	NA	NA

Terminal failures reveal class- and source-specific rejection mechanisms. Figure 8 reports the non-exclusive rates. Random recipes fail statistical and spectral gates much more often than cached LLM recipes. Healthy LLM candidates are rejected most often for unsupported envelope evidence, whereas fault candidates more often violate the robust statistical domain. A slot may fail several gates, so the percentages are non-additive. The same report supplies the next feedback round with each measured violation and its bound. Closed-loop admission therefore turns calibrated evidence into an auditable pool-construction decision rather than a post hoc quality score.

5.6. Verifier components support the frozen PU pool rather than only rejecting candidates

The cached replay directly separates the gates from the LLM proposal source. Disabling M2, M3, or M4 expands slot admission, while disabling M5 leaves the per-slot decision unchanged and alters only pool-level diversity selection. Each single-gate removal produces negative paired Accuracy and Macro-F1 differences from the complete gate at every

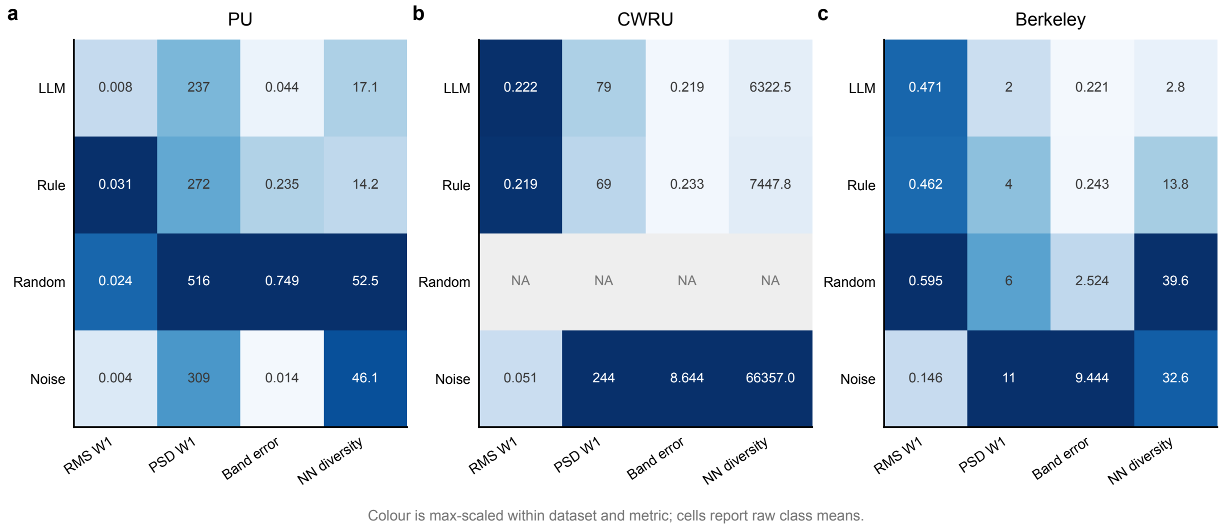

Figure 6: Class-averaged physical diagnostics on PU, CWRU, and Berkeley. Cells contain raw RMS Wasserstein-1, PSD-CDF Wasserstein-1, relative band-energy error, and nearest-neighbour diversity values; colour is scaled within each dataset and metric only. NA denotes a pool that does not exist in the frozen archive, not an imputed value. Diversity has no universally optimal direction.

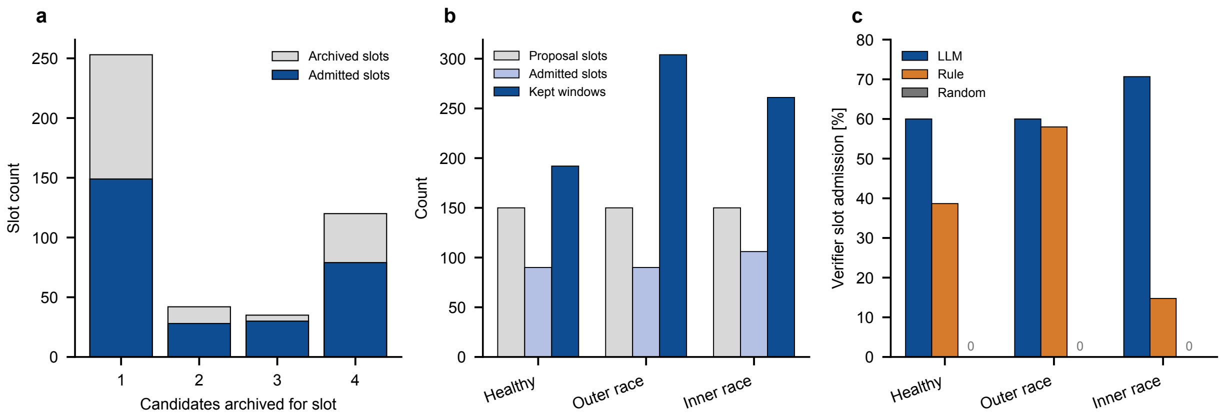

Figure 7: PU proposal accounting. (a) Number of archived candidates per proposal slot. (b) One-to-many mapping from 150 class slots to admitted slots and diversity-kept windows. (c) Verifier slot-admission rates for LLM, rule, and random recipes. Random open-loop remains a downstream control; random recipes subjected to the verifier produce no balanced admitted pool.

evaluated shot count (Table 10). M3 removal worsens the PSD diagnostic, whereas M4 removal can lower that one distance while still reducing downstream utility; thus no individual physical metric is treated as a universal quality score.

The $0.5\delta_y$ pool is byte-identical to the no-M5 pool under the fixed selection seed, so its downstream rows are an exact equivalence rather than a second independent training run. The reference threshold reproduces the frozen full pool. At $2.0\delta_y$, the cached healthy candidates cannot form the predeclared balanced budget, and no reduced-budget comparison is substituted.

Table 9
Per-class synthetic-to-real audit for the exact physics-v3 pool selections. NRMSE uses class-reference per-channel standardization; $|\rho|_{\max}$ is global-energy-normalized multichannel cross-correlation maximized over all linear lags. Exact denotes byte-identical float32 windows. No distance or correlation threshold is used to infer copying. Berkeley is exemplar-background simulation, so its similarity values characterize retained carrier structure rather than independent waveform generation.

dataset	pool	class	synthetic/reference	exact	min NRMSE	max $\lvert\rho\rvert$
PU	llm	healthy	150/1200	0	0.405	0.998
	llm	OR	150/1202	0	0.348	0.998
	llm	IR	150/1444	0	0.657	0.997
	rule	healthy	150/1200	0	0.500	0.997
	rule	OR	150/1202	0	0.440	0.997
	rule	IR	150/1444	0	0.744	0.997
	random_open_loop	healthy	150/1200	0	0.428	0.996
	random_open_loop	OR	150/1202	0	0.353	0.992
	random_open_loop	IR	150/1444	0	0.574	0.997
	noise_aug	healthy	150/1200	0	0.032	1.000
	noise_aug	OR	150/1202	0	0.028	1.000
	noise_aug	IR	150/1444	0	0.032	1.000
CWRU	llm	healthy	20/83	0	1.284	0.300
	llm	IR	20/163	0	0.650	0.281
	llm	B	20/163	0	0.149	0.252
	llm	OR	20/287	0	0.788	0.301
	rule	healthy	20/83	0	0.925	0.671
	rule	IR	20/163	0	0.610	0.320
	rule	B	20/163	0	0.142	0.240
	rule	OR	20/287	0	0.775	0.347
	random_open_loop	all	–	–	–	–
	noise_aug	healthy	20/83	0	0.260	0.972
	noise_aug	IR	20/163	0	0.037	1.000
	noise_aug	B	20/163	0	0.018	1.000
	noise_aug	OR	20/287	0	0.038	1.000
Berkeley	llm	healthy	20/629	0	0.310	0.932
	llm	degraded	20/1207	0	0.425	0.989
	rule	healthy	20/629	0	0.333	0.969
	rule	degraded	20/1207	0	0.337	0.999
	random_open_loop	healthy	20/629	0	3.925	0.826
	random_open_loop	degraded	20/1207	0	1.494	0.806
	noise_aug	healthy	20/629	0	0.043	0.999
	noise_aug	degraded	20/1207	0	0.035	0.999

Table 10
PU verifier-gate ablation from the cached 450-slot/class $K = 3$ archive. Each evaluable condition uses the frozen file split, 150 synthetic windows/class, and 20 paired CNN seeds. Entries are mean downstream deltas (Accuracy/Macro-F1) from the full gate; PSD-W1 is class-averaged. $2.0\delta_y$ cannot form the fixed healthy-class budget and is not evaluated at a reduced budget.

setting	slot accept.	PSD-W1	$n = 5$ ΔA/ΔF1	$n = 10$ ΔA/ΔF1	$n = 25$ ΔA/ΔF1	status
full gate	0.636	236.9	+0.000/+0.000	+0.000/+0.000	+0.000/+0.000	reference
without M2	0.824	242.1	-0.005/-0.010	-0.012/-0.014	-0.003/-0.004	B=150/class
without M3	0.684	263.5	-0.014/-0.014	-0.012/-0.013	-0.001/-0.002	B=150/class
without M4	0.744	225.0	-0.016/-0.018	-0.012/-0.014	-0.010/-0.011	B=150/class
without M5	0.636	228.0	-0.019/-0.021	-0.013/-0.015	-0.009/-0.010	B=150/class
$0.5\delta_y$	0.636	228.0	-0.019/-0.021	-0.013/-0.015	-0.009/-0.010	same as no M5
$1.0\delta_y$	0.636	236.9	+0.000/+0.000	+0.000/+0.000	+0.000/+0.000	same pool
$2.0\delta_y$	0.636	–	–	–	–	healthy 64/150

6. Cross-condition evidence and negative results

The root-cause framework also predicts where BREEZE should fail. Explicit recipes remove target distribution fitting from generator weights, but they do not identify an unknown mapping from operating condition to fault morphology. The following stress tests probe that missing knowledge, and their protocol value is to prevent extrapolation from positive within-condition results.

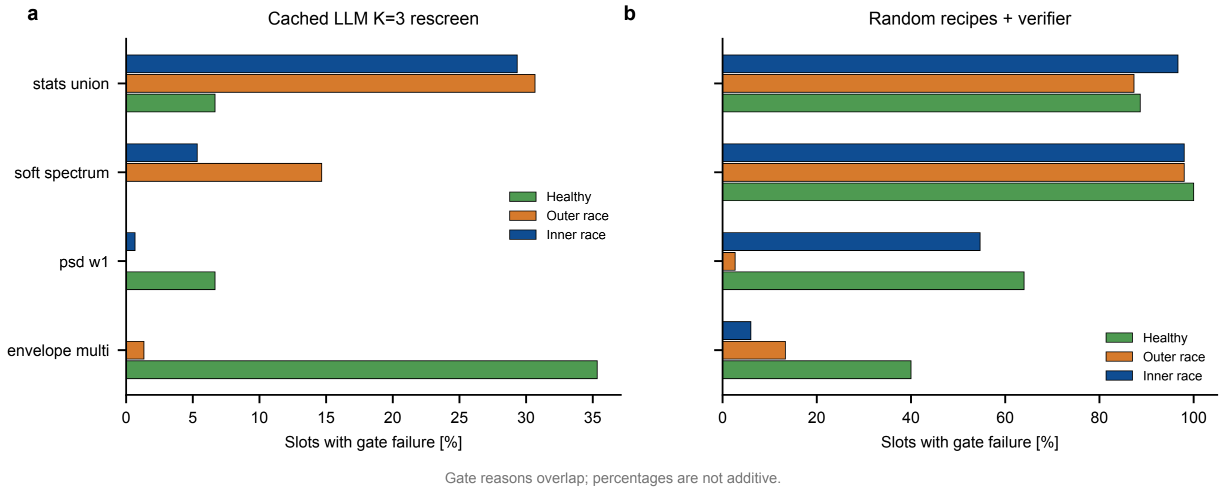

Figure 8: Non-exclusive PU gate-failure rates by class and recipe source. (a) Cached LLM $K = 3$ rescreen. (b) Random recipes passed through the same verifier. Statistics, soft-spectrum, PSD-W1, and multi-band envelope failures are computed from frozen slot reports.

6.1. CWRU source-load transfer succeeds while PU morphology does not transfer

CWRU source-load0 recipes transfer to the tested loads 1–3, whereas the PU LOCO protocol remains unsuccessful. Figure 9 places these outcomes side by side: PU v1 fails 57/96 registered cells, and v2 fails 60/96. The PU failures cluster by held-out condition and persist across Accuracy, Macro-F1, and shot counts, locating the unresolved boundary in condition-dependent morphology.

The six-stage PU audit localizes the missing morphology–condition map from kinematics through source-only evidence (Fig. 10). v1 exposed a kinematic mismatch, and v2 corrected frequency handling while downstream performance remained unsuccessful. v3 stopped when train-condition morphology mapping could not form an admissible balanced pool. v4 stopped after both train-only candidate-pool strategies failed admission, and v5 showed that its candidate criterion also admitted wrong-label/noise controls. v6 stopped because source-only cyclostationary coherence failed to separate all four held-out targets. The formal held-out evaluations are v1 and v2; v3–v6 remain development or admission stops.

6.2. Incomplete protocol metadata stops broader milling claims

The core milling evidence is limited to datasets with complete, auditable protocol metadata. The private machine-tool dataset used in earlier drafts therefore leaves the core evidence. Its speed, geometry, file-to-operation manifest, and audited physical label mapping are incomplete, which prevents a public reproducible portability claim.

UMich and MU-TCM define two distinct protocol stops. In the available UMich protocol, experimental condition and process metadata are confounded with wear labels, allowing a classifier to exploit condition identity. The current files leave these effects entangled, so a valid formal test requires repeated conditions or a revised acquisition design. MU-TCM v3 stops at train-only inner validation, where only 2/6 predeclared comparisons pass; preregistration and formal held-out evaluation were therefore withheld. Both datasets remain boundary evidence.

7. Discussion

7.1. Component tests answer the three evaluation questions

The source ablations answer the first question by establishing an LLM recipe contribution within the controlled BREEZE pipeline. On PU, random recipes subjected to the verifier produce 0 admitted slots and therefore constitute a capacity failure: no balanced downstream pool exists for an effectiveness comparison. The distinct random open-loop control is substantially weaker downstream. LLM recipes also exceed the fixed rule source in every registered PU cell and every provenance-valid CWRU family.

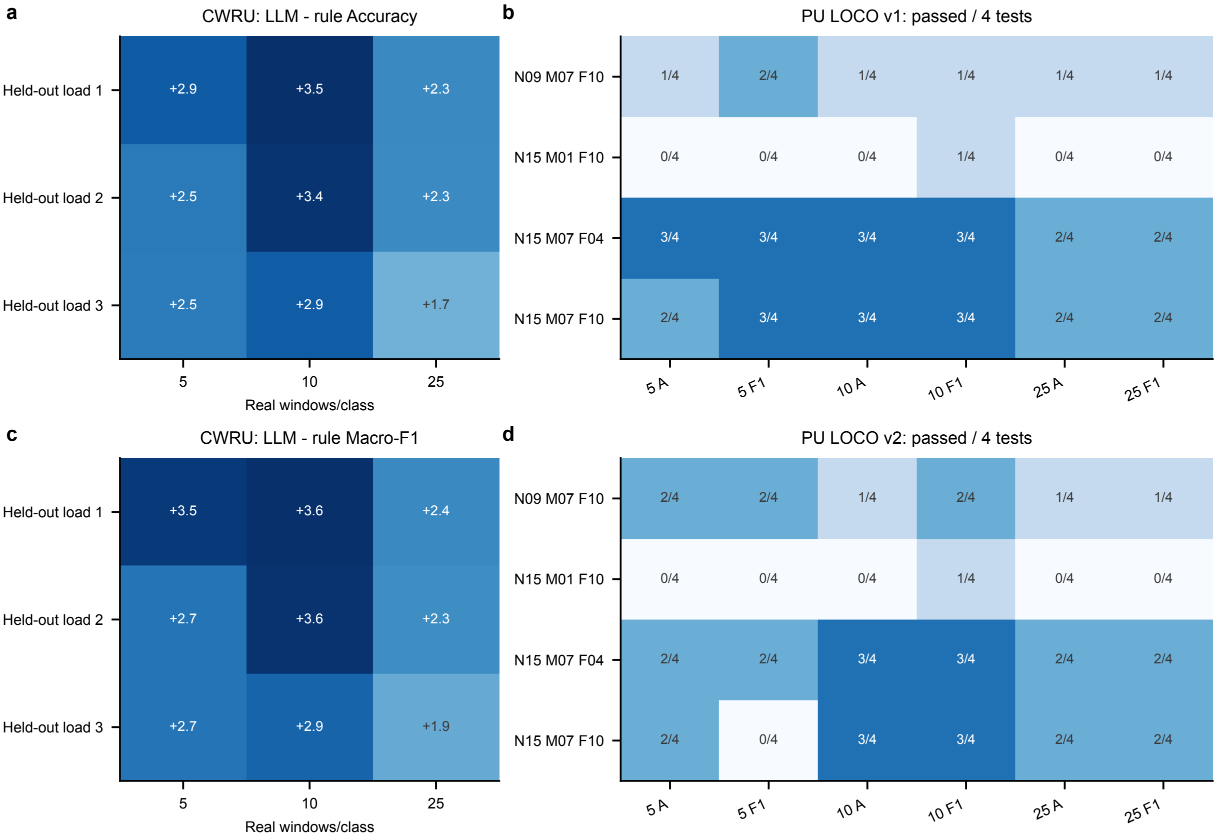

Figure 9: Cross-condition evidence. (a,c) CWRU LLM–rule Accuracy and Macro-F1 deltas when load0-derived pools are evaluated on held-out loads 1–3. The archived held-out-load0 fold is excluded for target-load provenance. (b,d) Number of passed PU LOCO tests out of four registered comparisons in each condition–shot–metric cell for v1 and v2.

The admission records answer the second question by showing how the verifier shapes feasible pools. Random recipes subjected to the verifier form no balanced pool, whereas accepted LLM pools exhibit waveform, envelope, PSD, band-energy, statistical-distance, and synthetic-to-real similarity records under the frozen diagnostics. Measured violations also drive the next bounded feedback round.

The downstream protocols answer the third question by locating both stable utility and its transfer boundary. LLM recipes improve the registered PU and provenance-valid CWRU comparisons and provide a qualified lower-shot Berkeley advantage. Their extra value converges with the rule recipe at ten Berkeley shots, while PU LOCO remains unsuccessful across six predeclared stages.

Together, these answers support the proposed knowledge-residence change: structured recipes and fixed equations propose signals, while calibrated evidence controls admission. The supported operating point requires few-shot real calibration and provides inspectable records without training a new target generator.

7.2. Knowledge residence guides the choice of generator

The available target evidence determines when the BREEZE operating point is appropriate. It fits deployments with few labelled target faults, no dedicated accelerator budget for generator optimization, and a requirement for sample-level audit records. The real outer-training split remains necessary for gate calibration, and the downstream diagnostic model still requires training.

A trained generator becomes a reasonable alternative when a representative target corpus and compute budget support distribution fitting, hyperparameter selection, and checkpoint selection, especially when important morphology

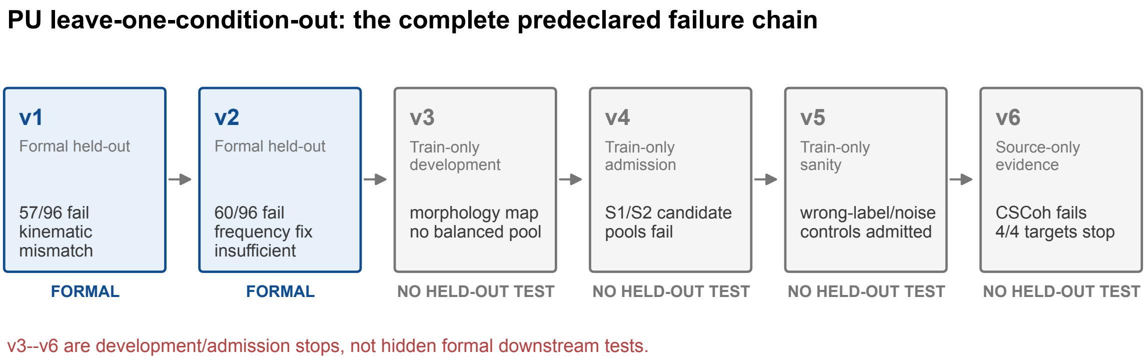

Figure 10: Complete PU LOCO failure chain. v1 and v2 are formal held-out evaluations; v3–v6 stop at train-only morphology, admission, sanity, or source-only evidence gates. The sequence prevents unsuccessful development stages from being reported as completed cross-condition tests.

cannot be expressed by the fixed renderer. The present evidence neither locates this crossover nor supports a benchmark ranking; the guidance follows the two paradigms' operational requirements.

7.3. Physical admission evidence versus physical truth

Train-calibrated gates provide auditable physical evidence within a finite set of observables. A candidate can satisfy RMS, PSD, and envelope gates while missing load-path, transfer-function, non-stationary, or degradation mechanisms; a rare real sample can likewise fall outside a finite training boundary. BREEZE therefore reports class-wise distances, downstream tests, and negative transfer results alongside gate pass rates. A pass means "admissible under the declared training-calibrated gates," not proof of membership in the full real distribution.

The rule comparison partially addresses the circularity created by shared renderer and verifier kinematics because rule and LLM recipes encounter the same renderer, physical checks, and downstream protocol. The random-plus-verifier audit also encounters those checks but stops at a zero-admission capacity failure; the separate random open-loop pool tests non-semantic recipe values without claiming verifier matching. These controls show that shared frequency definitions alone do not explain the useful LLM pool, but they cannot remove the circularity. External expert scoring, observables absent from the prompt, and new condition-disjoint datasets are required for a stronger physical-validity claim.

7.4. Reproducibility and remaining blockers

The repository supports replay from archived recipes through reported statistics. It stores split manifests, prompts, recipe JSON, local renderer seeds, waveform hashes, per-gate reports, pool manifests, seed-level classifier rows, registered tests, and table-generation assertions. Long-running pipelines are append-only or checkpoint-resumable. Frozen CSV/NPZ artifacts generate every reported number, and NA explicitly marks an unavailable pool.

Two provenance and baseline blockers remain. The original PU archive omits the provider version and provider-side sampling seed, so exact replay begins from the archived recipes instead of the original LLM response. Completed formal TimeGAN/DDPM and additional diagnostic-backbone comparisons are also unavailable, which keeps state-of-the-art and universal classifier claims outside the supported scope. Future work should complete those matched baselines, repeat generation to support independent-pool inference, test at least one non-CNN diagnostic backbone, evaluate cross-specimen and cross-machine splits, and preregister a new PU cross-condition representation before accessing held-out outcomes. The present seed-level tests quantify repeated few-shot/CNN variability around one shared fixed pool per method and do not estimate between-pool generation variability.

8. Conclusion

BREEZE provides training-free, auditable augmentation by using an LLM to select an executable recipe from supplied machine context, a fixed renderer to construct the waveform, and train-calibrated physical gates to control admission through bounded feedback and traceable records.

Across the frozen public protocols, this design isolates an LLM recipe contribution on PU, improves the registered CWRU comparisons, and provides a qualified lower-shot Berkeley benefit. The admitted pools also expose class-conditional physical diagnostics, proposal yield, and terminal failure mechanisms.

The supported scope is LLM-mediated recipe generation with physics-guided admission for three public frozen few-shot protocols. It does not establish state-of-the-art performance, superiority to trained generators, zero-real-data diagnosis, formal physical correctness, or cross-specimen and cross-machine generalization.

Data and code availability

PU and CWRU are public benchmarks, and Berkeley milling is available through the NASA prognostics repository subject to its terms. The Git repository contains the prompt builder and schema, deterministic renderer, verifier, table/figure scripts, frozen seed-level statistical outputs, PU pool arrays and manifests, physics manifests with SHA-256 hashes, and the evidence ledger. Primary CWRU and Berkeley waveform pools, recipes, and detailed candidate records currently reside in ignored `breeze/runs/` paths and are not part of the versioned Git tree. They will be deposited as a checksum-indexed repository release or data-repository supplement for exact waveform replay; until that deposit, the Git repository supports numerical-result audit and PU pool replay, but not complete CWRU/Berkeley waveform replay. API credentials, provider-side sampling state, and raw third-party datasets remain outside the release.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI and AI-assisted technologies

During preparation of this work, the authors used OpenAI Codex for code inspection, experiment orchestration, figure and table assembly, and language editing. The authors reviewed the evidence, verified the reported results, and take responsibility for the manuscript.

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